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Device tip

Abstract

In one aspect of the present disclosure, a device may include a shaft having a distal end, and a tip at the distal end of the shaft. The tip may include an opening defined by a surface of the tip. An instrument inserted through the shaft may extend distally out of the opening. The device also may include an elevator for engaging the instrument. The elevator may include a proximal end fixed relative to the surface of the tip, a proximal portion extending distally from the proximal end, and a distal portion extending distally from the proximal portion. A force exerted on the elevator bends the proximal portion to deflect the distal portion without bending of the distal portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This patent application is a continuation of and claims the benefit of priority to U.S. Nonprovisional patent application Ser. No. 17/724,284, filed on Apr. 19, 2022, which is a continuation of U.S. Nonprovisional patent application Ser. No. 15/909,205, filed on Mar. 1, 2018, which claims the benefit under 35 U.S.C. § 119 to U.S. Provisional Application No. 62/466,657, filed on Mar. 3, 2017, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) Various aspects of the present disclosure relate generally to tips for devices. More specifically, the present disclosure relates to endoscope tips for positioning endoscopic instruments.

BACKGROUND

(2) Endoscopic retrograde cholangiopancreatography (ERCP) is a technique to treat patients with pancreato-biliary disorders. In some instances, ERCP is performed using a side viewing endoscope. A tip of the side viewing endoscope may include side viewing optics, an optics lens wash, a side viewing light source, a side exit working channel, and an elevator assembly for elevating an instrument extending out of the side exit working channel. These features may aid a user in cannulating a subject's papilla, and accessing a target site in the subject's pancreas, with instruments for diagnostic and/or therapeutic procedures. However, elevator assemblies may have complex designs with a multitude of components, making them expensive to manufacture and maintain (e.g., clean and/or repair). Reducing the complexity of an elevator assembly may reduce

overall costs associated with using an endoscope. Moreover, when overall costs fall to a certain threshold, it may be practical for the endoscope to be used as a single-use endoscope. By using the endoscope only once before discarding it, cost and time burdens associated with maintaining the endoscope may be eliminated, and the risk of exposing a subject to infection through inadequate cleaning of the endoscope also may be eliminated, leading to improved outcomes.

SUMMARY

(3) Aspects of the present disclosure relate to, among other things, tips for devices. Each of the aspects disclosed herein may include one or more of the features described in connection with any of the other disclosed aspects.

(4) In one aspect of the present disclosure, a device may include a shaft having a distal end, and a tip at the distal end of the shaft. The tip may include an opening defined by a surface of the tip. An instrument inserted through the shaft may extend distally out of the opening. The device also may include an elevator for engaging the instrument. The elevator may include a proximal end fixed relative to the surface of the tip, a proximal portion extending distally from the proximal end, and a distal portion extending distally from the proximal portion. A force exerted on the elevator bends the proximal portion to deflect the distal portion without bending of the distal portion.

(5) According to another aspect of the present disclosure, the device may include one or more of the features below. The distal portion may have a greater cross-sectional area than the proximal portion. The distal portion may be thicker than the proximal portion along an anterior-posterior direction, the anterior-posterior direction being perpendicular to a proximal-distal direction. The tip may be comprised of a single material. The elevator may be comprised of a single material. An outer surface of the elevator and the surface of the tip may be portions of a continuous surface. An outer surface of the distal portion of the elevator may be continuous with the outer surface of the proximal portion of the elevator. The proximal portion may have a substantially straight rest configuration and a bent configuration. The proximal portion may move the distal portion through an arcuate path in an anterior direction and a proximal direction as the proximal portion moves from the rest configuration to the bent configuration.

(6) In another aspect of the present disclosure, a device may include a shaft having a distal end. The device also may include a tip at the distal end of the shaft. The tip may include an instrument opening. An instrument inserted through the shaft may extend distally out of the instrument opening. The tip also may include an elevator for engaging the instrument as the instrument extends distally out of the instrument opening. The elevator may include a passage extending therethrough. The device also may include a control wire coupled to the elevator. A proximal pulling force exerted on the control wire may deflect the elevator. The device also may include a first control wire opening and a second control wire opening. A first portion of the control wire may extend from the first control wire opening to the passage. A second portion of the control wire may extend from the second control wire opening to the passage. A third portion of the control wire may extend through the passage to link the first portion of the control wire to the second portion of the control wire.

(7) According to another aspect of the present disclosure, the device may include one or more of the features below. A handle at a proximal end of the shaft. The handle may include an actuation mechanism for exerting the proximal pulling force on the control wire. The actuation mechanism may be operatively coupled to the first portion of the control wire and the second portion of the control wire to exert the proximal pulling force simultaneously on the first portion of the control wire and the second portion of the control wire. The elevator may include opposing side surfaces. At least one of the side surfaces may include a channel for receiving at least a portion of the control wire. An opening into the passage may be in the channel. The tip may include sidewalls on opposing sides of the elevator. At least one of the sidewalls may include a channel for receiving at least a portion of the control wire. The tip may be a one-piece, monolithic, polymer component.

(8) In another aspect of the present disclosure, a device may include a shaft having a distal end. The

device also may include a tip at the distal end of the shaft. The tip may include an elevator. An instrument inserted through the shaft may extend distally to engage the elevator. The device also may include an actuator to deflect the elevator. The elevator may include a proximal portion having a proximal end fixed relative to the shaft. The elevator also may include a distal portion extending distally from the proximal portion. The proximal portion may have a smaller cross-sectional area than the distal portion, such that a force exerted on the elevator may bend the proximal portion to deflect the distal portion. The elevator may be made of a single, continuous piece of material.

(9) According to another aspect of the present disclosure, the device may include one or more of the features below. The single, continuous piece of material may be a single, continuous piece of injection-molded polymer. The proximal portion may include one or more indentations, such that the proximal portion may have one or more thin regions and one or more thick regions, to facilitate bending of the proximal portion along the one or more thin regions. The tip may further include at least one portion coupled to the single, continuous piece of material. The at least one portion may be made of a more rigid material than the single, continuous piece of material. The actuator may include a control wire coupled to the elevator.

(10) It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of any claimed inventions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate aspects of the present disclosure and together with the description, serve to explain the principles of the disclosure.
- (2) FIG. 1 shows an endoscope, in accordance with aspects of the present disclosure.
- (3) FIG. 2 shows a perspective view of a tip of the endoscope of FIG. 1, in accordance with aspects of the present disclosure.
- (4) FIG. 3 shows a front view of the tip of the endoscope of FIG. 1, in accordance with aspects of the present disclosure.
- (5) FIG. 4 shows a side view of the tip of the endoscope of FIG. 1 in a first configuration, in accordance with aspects of the present disclosure.
- (6) FIG. 5 shows a side view of the tip of the endoscope of FIG. 1 in a second configuration, in accordance with aspects of the present disclosure.
- (7) FIG. 6 shows a front view of an alternative endoscope tip, in accordance with aspects of the present disclosure.
- (8) FIGS. 7A and 7B show top and side views of a living hinge of an endoscope tip, in accordance with aspects of the present disclosure.
- (9) FIGS. 8A and 8B show top and front views of an elevator, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(10) The present disclosure is drawn generally to tips for devices, and more specifically to endoscope tips for positioning endoscopic instruments. Reference now will be made in detail to aspects of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. The term “distal” refers to a portion farthest away from a user when introducing an instrument into a subject. By contrast, the term “proximal” refers to a portion closest to the user when placing the instrument into the subject. The terms “anterior” and “posterior” refer to directions and/or regions extending perpendicular to proximal and distal directions. Though the following description refers to “endoscope” or “endoscopy,” the principles/aspects described herein

may be used with any suitable introduction sheath or device, even if such sheath or device fails to include one or more features typically associated with “endoscopes.” It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features claimed. Further, as used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not necessarily include only those elements, but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. The term “exemplary” is used in the sense of “example,” rather than “ideal.”

(11) FIG. 1 shows a device **100** for positioning an instrument **102**. Device **100** may be any type of endoscope, such as a duodenoscope. Instrument **102** may include, for example, a guidewire, cutting or grasping forceps, a biopsy device, a snare loop, an injection needle, a cutting blade, scissors, a retractable basket, a retrieval device, an ablation and/or electrophysiology catheter, a stent placement device, a surgical stapling device, a balloon catheter, a laser-emitting device, and/or any other suitable instrument. In one example, device **100** may facilitate cannulation of a subject's papilla, and placement of instrument **102** therein, during an ERCP procedure.

(12) Device **100** may include a shaft **104**. Shaft **104** may include a tube **106** that may be sufficiently flexible so as to have the ability to bend, rotate, and/or twist while being inserted into and/or through a subject's tortuous anatomy to a target site in the subject's body. Shaft **104** may have one or more lumens (not shown) extending therethrough. The lumens may include, for example, control wire lumens for receiving one or more control wires, a working lumen for receiving instrument **102**, a fluid lumen for delivering a fluid, an illumination lumen for receiving at least a portion of an illumination assembly (not shown), and/or an imaging lumen for receiving at least a portion of an imaging assembly (not shown).

(13) Device **100** also may include a tip **108** at a distal end of shaft **104**. Tip **108** may be attached to the distal end of shaft **104**. For example, tip **108** may be a cap configured to receive the distal end of shaft **104**. Tip **108** may include one or more openings that communication with the lumens of shaft **104**. For example, tip **108** may include control wire openings **110** and **112** through which one or more control wires may exit from the control wire lumens of shaft **104**, a working opening **114** through which instrument **102** may exit from the working lumen of shaft **104**, a fluid opening or nozzle **116** through which fluid may be emitted from the fluid lumen of shaft **104**, an illumination opening or window **118** through which light may be emitted, and/or an imaging opening or window **120** for receiving light used by the imaging device to generate an image. While not depicted in FIG. 1, it should be understood that illumination opening **118** may include a light emitter (e.g., a light-emitting diode, or the like) coupled to a cable or wire running through the illumination lumen of shaft **104**, and/or imaging opening **120** may include an imager (e.g., a charge-coupled device image sensor, a complementary metal-oxide image semiconductor, or the like) coupled to a cable or wire running through the imaging lumen of shaft **104**.

(14) Control wire openings **110** and **112** may be on a distal-facing surface **124** of tip **108**. Working opening **114** may be on a distal-facing surface **126** of tip **108**. Distal-facing surface **126** may be recessed from distal-facing surface **124** in a proximal direction. Illumination opening **118** and imaging opening **120** may be on an anterior-facing surface **128**. Fluid opening **116** may be on distal-facing surface **124**, and may open toward illumination opening **118** and imaging opening **120** to, for example, direct a stream of rinsing fluid across illumination opening **118** and/or imaging opening **120** to wash off bodily fluids or debris.

(15) Working opening **114** may open into a cavity **130** of tip **108**. Cavity **130** may be defined by distal-facing surface **126**, lateral-facing sidewalls **132** and **134**, a posterior-facing surface **136**, and an anterior-facing surface **138** of an elevator **140**. Instrument **102** may extend distally out of working opening **114** and into cavity **130**, where instrument **102** may be positioned for engagement with anterior-facing surface **138** of elevator **140**. Anterior-facing surface **138** of elevator **140** may

include, for example, an indentation **168** (e.g., a slot, groove, cavity, or the like) formed therein. Indentation **168** may receive at least a portion of instrument **102** to exert a holding or gripping force on instrument **102**.

(16) Elevator **140** may be cantilevered, having a fixed proximal end **142** at distal-facing surface **126**, and a free distal end **144**. Elevator **140** may include a proximal portion **146** and a distal portion **148**. Proximal portion **146** may be thinner than distal portion **148** to facilitate bending of proximal portion **146**. This is because distal portion **148**, due to its thickness, may be more rigid than proximal portion **146**. In some examples, proximal portion **146** may bend while distal portion **148** may remain unbent.

(17) Proximal portion **146** may have a rectangular cross-sectional shape to facilitate bending along the anterior-posterior direction. Additionally or alternatively, a length of proximal portion **146** (along the proximal-distal direction) may be greater than a width of proximal portion **146** (along a lateral direction perpendicular to the proximal-distal direction), and the width of proximal portion **146** may be greater than a height/thickness of proximal portion **146** (along the anterior-posterior direction). Additionally or alternatively, proximal portion **146** may have a substantially constant thickness all the way from distal-facing surface **126** to a proximal end of distal portion **148**, where elevator **140** may undergo an abrupt increase in thickness. Anterior-facing surface **138** may be angled (e.g., inclined) relative to proximal portion **146** to facilitate engagement between anterior-facing surface **138** and instrument **102**. Additionally or alternatively, one or more of a posterior-facing surface of proximal portion **146** and an anterior-facing surface of proximal portion **146** may be substantially planar and/or substantially perpendicular to one or more of lateral-facing sidewalls **132** and **134**.

(18) Proximal portion **146** may act as a living hinge to allow distal portion **148** of elevator **140** to move relative to the rest of tip **108**. Proximal portion **146** may bend such that distal portion **148** may travel along an arcuate path that may cover a distance in a proximal-distal direction and an anterior-posterior direction. In some examples, elevator **140** may be integral with the rest of tip **108**. For example, tip **108**, as shown in FIG. 2, may be a single, monolithic, injection-molded component. Tip **108** may be made of a polymeric material, such as polyethylene, polypropylene, and/or any other suitable polymer. Tip **108** may be made by any other suitable process, and with any other suitable material, for forming tip **108** as a single piece of the same material throughout.

(19) Distal portion **148** may be thicker than proximal portion **146** to provide space for a passage **150**. The thickness of distal portion **148** may ensure that distal portion **148** does not bend, and that bending of elevator **140** is limited to proximal portion **146**. Passage **150** may be a through-hole extending between opposing lateral-facing surfaces **152** and **154** of elevator **140**. Passage **150** may receive control wire **122**. For example, a first portion **156** of control wire **122** may extend distally from control wire opening **110** to lateral-facing surface **152** and passage **150** (see FIG. 3). A second portion **158** of control wire **122** may extend distally from control wire opening **112** to lateral-facing surface **154** and passage **150**. A third portion (not shown) of control wire **122** may extend through passage **150**. First portion **156**, second portion **158**, and the third portion of control wire **122** may form a U-shaped extent of control wire **122**.

(20) Channels **160**, **162**, **164**, and **166** may be provided in sidewall **132**, lateral-facing surface **152**, lateral-facing surface **154**, and sidewall **134** for receiving portions of control wire **122**. The received portions of control wire **122** may slide along channels **160** and **166** as control wire **122** is retracted into and/or extended out of control wire openings **110** and **112**. In the depicted example, passage **150** may open into channels **162** and **164**. Control wire **122** may be strung through passage **150** by deflecting elevator **140** in a posterior direction to expose passage **150** from behind sidewalls **132** and **134**, inserting control wire **122** through passage **150**, aligning control wire **122** with channels **160**, **162**, **164**, and **166**, inserting control wire **122** into control wire openings **110** and **112**.

(21) By pulling one or both of first portion **156** and second portion **158** of control wire **122** in a proximal direction, the user may retract control wire **122** into control wire openings **110** and **112**.

This retraction may cause proximal portion **146** of elevator **140** to bend as distal portion **148** of elevator **140** is drawn toward distal-facing surface **124**. FIGS. **3** and **4** show views of tip **108** where elevator **140** is at rest, while FIG. **5** shows a view of tip **108** where proximal portion **146** of elevator **140** is bent due to forces exerted on distal portion **148** of elevator **140** by control wire **122**. Pulling both of first portion **156** and second portion **158**, which are on opposite sides of elevator **140**, may ensure that elevator **140** may be drawn toward distal-facing surface **124** with little or no twisting or torqueing, to ensure that instrument **102** is brought into a desired position.

(22) Instrument **102** may be held in indentation **168** of distal portion **148** as distal portion **148** is drawn toward distal-facing surface **124**. As distal portion **148** is deflected, distal portion **148** may deflect instrument **102**, thereby adjusting the position of instrument **102**. Instrument **102** may be extended out of, retracted into, and/or rotated (e.g., twisted) in the working lumen of shaft **104** and working opening **114** before, during, and/or after instrument **102** is deflected by distal portion **148**. Continued pulling of control wire **122** may cause distal portion **148** to pin instrument **102** against a protrusion **170** on distal-facing surface **124**. This may lock instrument **102** in place relative to tip **108**. When locked, the ability to extend, retract, and/or rotate instrument **102** may be hindered or eliminated. In one example, protrusion **170** may have a shape complementary to a shape of indentation **168** to facilitate holding or locking of instrument **102** between protrusion **170** and distal portion **148**. For example, protrusion **170** may have a wedge-like shape. When the user ceases to pull control wire **122**, and/or extends control wire **122** distally, proximal portion **146** may inherently seek to move to its unbent state, thereby moving distal portion **148** away from distal-facing surface **124** and back toward its orientation as shown in FIGS. **2-4**. This may unlock instrument **102**.

(23) Device **100** also may include a handle **172** coupled to a proximal end of shaft **104**. Handle **172** may include one or more lumens (not shown) that communicate with the lumen(s) of shaft **104**. Handle **172** also may include one or more ports **174** that open into the one or more lumens of handle **172**. For example, instrument **102** may be inserted into port **174** on its way to the working lumen of shaft **104**. In one example, handle **172** may include a cable **176**. Cable **176** may couple one or more external systems (not shown) to device **100**. Cable **176** may, for example, couple handle **172** to a fluid supply, an illumination source, and/or an imaging processor or display.

(24) Handle **172** also may include an actuation mechanism **178**. Actuation mechanism **178** may include one or more knobs, buttons, levers, switches, and/or any other suitable actuators, for controlling at least one of deflection of shaft **104**, the delivery of fluid, the emission of illumination, and/or imaging functions. In one example, actuation mechanism **178** may be operatively coupled to proximal portions (not shown) of control wire **122** that extend proximally from control wire openings **110** and **112** into handle **172**. The user may manipulate actuation mechanism **178** to selectively exert at least one of a pulling force and a pushing force on control wire **122** to control the position of elevator **140**, and thereby control the position of instrument **102**.

(25) FIG. **6** shows an alternative embodiment of a tip **108'**. Tip **108'** may be similar to tip **108**, except tip **108'** may be made of separate components coupled together at a joint **109'**. In one example, portions of tip **108'** on a first side of joint **109'** may be a single, monolithic component, while portions of tip **108'** on a second side of joint **109'** may be another single, monolithic component. By producing the components separately and then joining them together, one material may be used for one of the components, while another material (with different properties) may be used for the other component. For example, the component to the left of joint **109'** may be made of a material that is more flexible than the material used to make the component to the right of joint **109'**, to facilitate bending of the elevator.

(26) In the example shown in FIG. **6**, the component to the left of joint **109'** may include an elevator, a working opening, control wire openings, a distal-facing surface, a lateral-facing sidewall, and/or a posterior-facing surface, similar to those of tip **108**. The component to the right of joint **109'** may include a fluid opening, a lateral-facing sidewall, an anterior-facing surface, an

imaging opening, and/or an illumination opening, similar to those of tip **108**. It should be understood, however, that joint **109**' may divide tip **108**' in any other suitable manner. It also is contemplated that joint **109**' may follow a straight line across tip **108**', or may include one or more bends and/or corners, as it crosses tip **108**'. It also is contemplated that joint **109**' may divide tip **108**' into more than two components.

(27) FIGS. 7A and 7B show views of an alternative embodiment of a proximal portion **146**' of an elevator. Proximal portion **146**' may have one or more elements thereon for precisely controlling bending of proximal portion **146**'. In one example, proximal portion **146**' may include cutouts or indentations **147**' (e.g., notches, grooves, slots, holes, and the like) to facilitate bending by creating thinner portions of proximal portion **146**'. Additionally or alternatively, proximal portion **146**' may include protrusions (e.g., ridges, spines, and the like) (not shown) to limit bending by creating thicker portions of proximal portion **146**'. The cutouts, indentations, and/or protrusions may be on a posterior-facing surface of proximal portion **146**', an anterior-facing surface of proximal portion **146**', and/or one or more lateral-facing surfaces of proximal portion **146**'.

(28) FIGS. 8A and 8B show views of an alternative embodiment of an elevator **140**' including a plurality of indentations **168**' and **169**'. Indentations **168**' and **169**' may have different characteristics. For example, one of indentations **168**' and **169**' may be wider, deeper, and/or longer than the other. Indentation **168**' may be sized to hold larger instruments, and indentation **169**' may be sized to hold smaller instruments.

(29) It will be apparent to those skilled in the art that various modifications and variations can be made in the disclosed systems and methods without departing from the scope of the disclosure. Other aspects of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the features disclosed herein. It is intended that the specification and examples be considered as exemplary only.

Claims

1. A device, comprising: a shaft; a distal tip including an interior sidewall and a first channel positioned on the interior sidewall; and an elevator including an exterior sidewall, a proximal portion that is fixed relative to the interior sidewall, a distal portion that is movable relative to the proximal portion, and a second channel positioned on the exterior sidewall; wherein, when the elevator moves to a first position, the first channel and the second channel are each configured to receive a portion of a control wire, and when the elevator moves to a second position the control wire is configured to move out of the first channel and remain inside the second channel; wherein the proximal portion bends relative to the distal portion when the elevator moves from the first position to the second position, wherein the distal tip includes a distally-facing surface defining a first control wire opening and a second control wire opening, wherein a first portion of the control wire extends from the first control wire opening, wherein a second portion of the control wire extends from the second control wire opening, and wherein a third portion of the control wire extends between the first portion and the second portion at the distal portion of the elevator.
2. The device of claim 1, wherein the shaft is configured to receive an instrument that extends distally out of the distal tip, and the elevator is configured to deflect the instrument relative to the distal tip when moving from the first position to the second position, thereby causing the control wire to move out of the first channel.
3. The device of claim 1, wherein the distal portion of the elevator includes a width defined between the exterior sidewall and an opposing exterior sidewall of the elevator.
4. The device of claim 3, wherein the elevator includes an opening positioned within the second channel and a passage that extends across the width of the distal portion and terminates at the opening.
5. The device of claim 4, wherein the elevator includes a second opening positioned on an opposing

exterior sidewall that is opposite of the exterior sidewall of the elevator, and the passage terminates at the second opening.

6. The device of claim 5, wherein the control wire received within the second channel enters the passage at the opening and exits the passage at the second opening.

7. The device of claim 4, wherein the passage has a length that is less than the width of the distal portion.

8. A device, comprising: a shaft; a distal tip including an interior surface and a first channel extending along the interior surface; and an elevator including an exterior surface, a proximal portion that is fixed relative to the interior surface of the distal tip, a distal portion that is movable relative to the proximal portion, and a second channel extending along the exterior surface; wherein, when the elevator moves to a lowered position, the first channel is configured to receive a proximal portion of a control wire and the second channel is configured to receive a distal portion of the control wire, and when the elevator moves to a raised position the proximal portion of the control wire is guided out of the first channel and the distal portion of the control wire is retained inside the second channel, wherein the proximal portion bends relative to the distal portion when the elevator moves from the lowered position to the raised position, wherein the distal tip includes a distally-facing surface defining a first control wire opening and a second control wire opening, wherein a first portion of the control wire extends from the first control wire opening, wherein a second portion of the control wire extends from the second control wire opening, and wherein a third portion of the control wire extends between the first portion and the second portion at the distal portion of the elevator.

9. The device of claim 8, wherein the first channel is at least partially open along the interior surface of the distal tip for receiving the proximal portion of the control wire, and the second channel is at least partially open to the exterior surface of the elevator for receiving the distal portion of the control wire.

10. The device of claim 8, wherein the control wire is configured to move the elevator to the raised position in response to the proximal portion moving proximally relative to the distal tip, such that an instrument received on the elevator is deflected relative to the distal tip.

11. The device of claim 8, wherein the control wire is configured to move the elevator to the lowered position in response to the proximal portion moving distally relative to the distal tip, such that an instrument received on the elevator is longitudinally aligned with the distal tip.

12. The device of claim 8, wherein the proximal portion of the control wire exits the first channel when the elevator moves from the lowered position to the raised position, and wherein the proximal portion of the control wire enters the first channel when the elevator moves from the raised position to the lowered position.

13. The device of claim 8, wherein the distal portion of the elevator includes an indentation, wherein the distally-facing surface of the distal tip includes a protrusion that is complementary to the indentation of the elevator, and wherein the protrusion is configured to face the distal portion of the elevator when in the raised position.

14. The device of claim 8, wherein the distal portion of the control wire remains fixed relative to the second channel when the elevator moves between the lowered position and the raised position.

15. A device, comprising: a shaft; a distal tip including an interior surface and a first channel on the interior surface; and an elevator including an outer surface, a proximal portion that is fixed relative to the interior surface, a distal portion that is movable relative to the proximal portion, and a second channel on the outer surface; wherein, when the elevator is undeflected relative to the distal tip, each of the first channel and the second channel are configured to receive a control wire, and when the elevator is deflected relative to the distal tip, the first channel is configured to release the control wire and the second channel is configured to hold the control wire; wherein the proximal portion bends relative to the distal portion when the elevator deflects relative to the distal tip, and wherein the distal tip includes a distally-facing surface defining a first control wire opening and a

second control wire opening, wherein a first portion of the control wire extends from the first control wire opening, wherein a second portion of the control wire extends from the second control wire opening, and wherein a third portion of the control wire extends between the first portion and the second portion at the distal portion of the elevator.

16. The device of claim 15, wherein the control wire moves out of the first channel when the control wire translates proximally relative to the shaft to deflect the elevator proximally relative to the distal tip, such that the control wire is removed from the first channel, and wherein the control wire remains inside the second channel when the control wire translates proximally relative to the shaft to deflect the elevator proximally relative to the distal tip, such that the control wire moves with the elevator.

17. The device of claim 15, wherein the distal portion of the elevator includes an indentation configured to receive a portion of an instrument therein, and wherein the distally-facing surface of the distal tip includes a protrusion that is complementary to the indentation of the elevator, wherein the protrusion is configured to face the distal portion of the elevator when deflected relative to the distal tip.

18. The device of claim 17, wherein the protrusion is positioned between the first control wire opening and the second control wire opening.

19. The device of claim 15, wherein the first portion, the second portion, and the third portion of the control wire form a U-shape.

20. The device of claim 15, wherein the elevator includes a third channel on a second exterior sidewall of the elevator, the first portion of the control wire extends from the first control wire opening to the second channel, and the second portion of the control wire extends from the second control wire opening to the third channel.
